

EEE 491 Lab Assignment

Lab-COMPARE: Feature Vectors Comparator

1. Technical Specifications/Requirements

Lab-COMPARE shall be operated in two modes: RECORD and COMPARE. In the RECORD mode, Lab-COMPARE shall treat feature vectors of a spoken number as reference and store them in a RAM called REF-RAM. The spoken number will be identified by 4-bit NUMBER input signal which determines the start address of the feature vectors of the spoken number in REF-RAM. REF-RAM shall contain all the feature vectors for all ten numbers. In the COMPARE mode, Lab-COMPARE shall store the feature vectors of a spoken number in a RAM called COMP-RAM, and then start to calculate Euler distances between COMP-RAM contents and REF-RAM for each number. Finally, Lab-COMPARE shall compare the Euler distances and shall decide for the best match of a number and display the number on a seven segment display on Basys-3 board. Use “Block Memory generator” building block from the Vivado “IP Catalog”.

Lab-COMPARE in FPGA shall have the following input/output (IO) signals.

- “reset_in” input control signal, which resets all the registers in the design.
- “clock_in” input clock signal of the 100MHz clock oscillator on the Basys-3 board.
- “dct_addr_out” output signal that addresses the read-only port of the dual-port RAM in Lab-DCT.
“dct_data_in” input signal from the read-only port of the dual-port RAM in Lab-DCT.
- “record_in” input control signal which indicates if the samples are to be used as reference feature vectors.
- “number_in” input control signal which indicates the number whose feature vectors are to be stored as the reference.
- “seg_a ... seg_g” output signals which are used for displaying the spoken number and the number to speak (when record signal is active) on 7-segment display.
- “start_in” input control signal, which starts to read feature vector from Lab-DCT. In case for the last frame, Lab-COMPARE starts to recognize the spoken number. This signal is active high asserted on a single clock pulse.
- “ready_in” output status signal which indicates that feature vector is read from the Lab-DCT and stored in dual-port RAM. In case for the last frame, it indicates that the recognized number is displayed on 7-segment display. This signal is active high and asserted low just after the start signal is asserted.

After verifying your Lab-COMPARE design by using your test-bench simulation results, integrate Lab-COMPARE with the Lab-PCB, ADC board, Lab-ADC, Lab-WINDOW, Lab-CTRL, Lab-FFT, Lab-MEL, Lab-DCT. Control and observe the process sequence using start_in signal of Lab-CTRL and ready_out signals of the components. On the Basys-3 board, use a button for start_in signaling, a LED for each ready_out signals of the components, 7-segment display for displaying the spoken number (0 to 9), a mode switch (RECORD/COMPARE) and 4 switches for NUMBER signal that defines the number whose feature vectors are to be calculated during the RECORD mode.

2. Demonstration

- Present your test bench simulation results: inputs, outputs (waveforms). Show that Lab-COMPARE stores feature vectors of the defined number in REF-MEM. Show that the system stores feature vectors of the spoken number in COMP-MEM. Show that the system compares the feature vectors and finds the best match.

- Configure your integrated design on Basys-3 board. Use start_in button and ready_out LEDs of the components to observe/verify process sequence. Use RECORD/COMPARE switch, NUMBER switch and 7-segment display. If you need you may add more LEDs for further observations.
- Prepare the Lab-COMPARE demonstration set up. Set the RECORD/COMPARE switch to RECORD mode, set the NUMBER switch, see the set number on the 7-segment display, speak the set number to the microphone, and then let the system store the feature vectors of the set number. Repeat this for the remaining 9 numbers. Then switch to COMPARE mode, speak a number to the microphone, and let the system recognize the spoken number and display it on 7-segment display.

3. Guidance

- Test bench may store predefined feature vectors of 10 numbers in a ROM (filled with data in COE file). Test bench stores feature vectors of a number to be compared in a ROM (filled with data in COE file). Test bench addresses the ROM so that at each "start_in" signal activation a set of feature vector is available (as Lab-DCT memory output) to be read by the Lab-COMPARE. There may be several test benches to test various functions of the Lab-COMPARE.
- There may be more than one feature vectors for per number in order to increase recognition performance. The sizes of the memories depend on the number of feature vectors which are determined during Lab-MATLAB assignment.
- Let F be the number of frames to be processed. Lab-CTRL sends the start signal F times to the Lab-COMPARE. Lab-CTRL must count the start signals and generate memory address in order to store the feature vectors data to either REF-RAM or COMP-RAM.
- You may initially verify your system by recording and then comparing feature vectors of sinusoid signals in 10 different frequencies.